Customer Journey & Intent Tracking System

Core Platform / Integration Technical Architecture & Implementation Requirement

STARVNT ENTERTAINMENT

Prepared for Keshab | Technical Lead + Core Platform Integration

MVP-first | Privacy-conscious | Secure | Modular | Scalable

Executive Summary

This document defines the technical direction for StarVnt's Customer Journey & Intent Tracking System. It is intended to connect first-party visitor behaviour, acquisition attribution, referral attribution, sales activity, Core Commerce events and revenue outcomes into one privacy-conscious intelligence layer.

The system must help authorized StarVnt teams understand the customer journey without becoming a surveillance system. Analytics observes activity inside StarVnt-owned digital properties; Core Commerce, Revenue, CRM and Referral remain authoritative for their own domains.

> *Core principle: Track customer behaviour inside StarVnt, understand intent, preserve acquisition and referral attribution, connect the journey to real commercial outcomes, and isolate every domain and every user role appropriately.*

High-Level Journey

```
Source / Referral -> Visitor -> Session -> Service Interest -> Enquiry -> Qualification -> Quote -> Vendor Match ->
Booking -> Payment -> Revenue
```

Operating Principle

Start with a simple MVP, make the contracts and boundaries correct, then add intelligence progressively. Future layers such as predictive intent, AI lead scoring, recommendations and customer lifetime value should sit on top of the same event and attribution foundation.

1. Primary Objective

1. Where a visitor/customer came from
2. Whether the acquisition was organic, paid, social, campaign or referral
3. Which StarVnt services they are interested in
4. Which pages, packages or portfolio items they engaged with
5. Whether they started an enquiry
6. Whether they completed an enquiry
7. Whether the lead was verified or qualified
8. Whether a quote was requested
9. Whether vendor matches were viewed or selected
10. Whether booking or payment was initiated
11. Whether the customer eventually booked
12. How strong the current purchase intent is
13. Where the customer dropped off
14. What Sales should do next
15. Whether the journey generated revenue
16. Which source, campaign or referral ultimately generated that revenue

The system should connect these signals while maintaining strict privacy and domain boundaries.

2. Core Journey Model

```
Acquisition
Source / Campaign / Referral

-> Anonymous Visitor
SV-VIS-XXXX

-> Session
SV-SES-XXXX
```

```
-> Behaviour
Page / Service / Package / Portfolio / CTA

-> Lead
SV-LEAD-XXXX

-> Qualification
New -> Verified -> Qualified

-> Commerce
Quote -> Vendor Match -> Booking -> Payment

-> Revenue
Booking Value -> StarVnt Revenue -> Referral Attribution / Commission
```

This model allows StarVnt to reconstruct a customer journey without duplicating the underlying business domains.

3. System Boundary

Domain	Owns
Analytics	Behavioural events, visitor/session tracking, attribution, journey reconstruction, service interest, intent calculation and analytics reporting
CRM / Revenue CRM	Lead, sales activity, lead status, follow-up and qualification
Core Commerce	Customer identity, Inquiry, Quote, Booking, Booking status and Payment status
Revenue	Financial attribution, revenue calculation, booking value and StarVnt revenue
Referral System	Referral relationship, referral code/link, referral partner, referral attribution and referral reward/commission logic

Analytics should connect these systems, not recreate them.

4. First-Party Visitor Identity

Create a first-party anonymous Visitor ID and a Session ID. Examples:

```
SV-VIS-8F42K1
SV-SES-20260825-00182
```

The Visitor ID remains anonymous until a legitimate identity association occurs. If the visitor submits an enquiry, creates an account, logs in or otherwise becomes identifiable, the anonymous journey may be associated with the appropriate Lead/Customer record where legally permitted and properly disclosed.

- Visitor IDs are identifiers, not authentication credentials.
- Do not expose internal identifiers unnecessarily.
- Do not use visitor IDs as security tokens.
- Do not allow clients, vendors or referral partners to enumerate internal IDs.

5. Identity Association Model

```
Visitor -> Session -> Lead -> Customer -> Inquiry -> Quote -> Booking -> Payment
```

```
SV-VIS-1234 -> SV-LEAD-5678 -> CRM Customer ID -> Inquiry ID -> Quote ID -> Booking ID -> Payment ID
```

Analytics should reference authoritative identifiers from Core/CRM rather than creating duplicate customer identities.

6. Session Tracking

- Session start
- Session end
- Landing page
- Entry source
- Referrer
- UTM source / medium / campaign / content / term
- Referral ID/code
- Device category
- Browser
- Approximate geography where appropriate
- Pages viewed
- Session duration
- Exit page

Avoid collecting unnecessary personal or sensitive information.

7. Attribution Architecture

Attribution is a core architectural requirement. Support Organic, Direct, Social, Paid, UTM campaigns, Referral links, Referral codes, StarVnt campaigns and other approved acquisition sources.

The system must preserve acquisition information from the earliest known source through the commercial lifecycle.

```
Instagram Campaign A -> Visitor -> Lead -> Quote -> Booking -> Revenue
Referral Partner X -> Referral Link -> Visitor -> Lead -> Booking -> Revenue -> Referral Commission
```

Attribution Model for MVP

- Preserve sufficient raw attribution data to support first-touch attribution.
- Preserve sufficient raw attribution data to support last-touch attribution.
- Support a primary attribution model.
- Do not build complex multi-touch attribution initially unless the existing architecture already supports it.
- Keep enough raw data so multi-touch attribution can be introduced later.

8. Attribution Persistence

Capture the relevant acquisition context when a visitor first arrives.

```
visitor_id
first_touch_source
first_touch_medium
first_touch_campaign
first_touch_referral
first_touch_timestamp
```

```
last_touch_source
last_touch_medium
last_touch_campaign
last_touch_referral
last_touch_timestamp
```

When the visitor becomes a lead, preserve the attribution reference. When the lead becomes a booking, preserve the relationship rather than attempting to reconstruct attribution from incomplete historical data.

```
Attribution -> Visitor -> Lead -> Inquiry -> Quote -> Booking -> Revenue
```

Avoid creating multiple competing attribution records for the same commercial outcome.

9. Referral System Integration

```
Referral Link / Code -> Visitor -> Session -> Lead -> Inquiry -> Quote -> Booking -> Payment -> Revenue
```

The system should support determining referral-generated visitors, referral-generated leads, referral-attributed bookings, booking value, StarVnt revenue, referral conversion and applicable commission/reward.

> *Referral attribution does not grant customer analytics access. A referral partner must receive only explicitly approved referral-related information.*

10. Standard Event Architecture

```
event_id
event_name
event_version
timestamp
visitor_id
session_id
lead_id
customer_id
page_path
referrer
utm_source
utm_medium
utm_campaign
utm_content
utm_term
referral_id
service_id
metadata
```

Not every event needs every field. Event schemas must be explicit, versioned, validated, documented and backward-compatible where possible. Do not let arbitrary uncontrolled JSON become the permanent analytics schema.

11. Event Categories

Navigation

```
page_view
service_view
portfolio_view
pricing_view
faq_view
```

Engagement

```
cta_click
whatsapp_click
```

```
phone_click
portfolio_click
package_view
compare_click
```

Lead Funnel

```
lead_form_started
lead_field_completed
lead_form_submitted
lead_form_abandoned
lead_verified
```

Commerce

```
quote_requested
quote_viewed
vendor_match_viewed
vendor_selected
booking_started
payment_started
payment_completed
booking_confirmed
```

Account

```
signup_started
signup_completed
login
logout
```

Future Commerce Events

```
inquiry.created
inquiry.qualified
quote.created
quote.updated
quote.accepted
quote.rejected
booking.created
booking.status_changed
booking.cancelled
booking.completed
payment.started
payment.completed
payment.failed
payment.refunded
event.completed
```

12. Behavioural vs Business Events

Maintain a clear distinction between client/browser-originated behavioural events and authoritative server-side business events.

Type	Examples	Trust Model
Behavioural	page_view, cta_click, package_view	May originate from browser/client and is treated as untrusted input
Business	inquiry.created, quote.accepted, booking.created, payment.completed, revenue.recorded	Should originate from authoritative server-side systems

13. Event Integrity

Browser-generated analytics events must always be treated as untrusted input. Do not use browser events as authoritative evidence for payment, booking, revenue, referral payout or financial settlement.

```
Browser -> payment_completed != Payment successful
Payment Gateway / Core Commerce -> trusted payment.completed -> Analytics
Core Commerce -> booking.created -> Analytics
```

14. Commerce Event Contracts

```
inquiry.created
quote.created
quote.accepted
booking.created
booking.status_changed
payment.completed
```

Each trusted event should contain stable references where applicable, such as event_id, event_name, event_version, timestamp, customer_id, lead_id, inquiry_id, quote_id, booking_id, payment_id, visitor_id, session_id, attribution_id and referral_id. The exact contract should be proposed after reviewing the existing Core Platform.

15. Event Ownership

Event	Source / Owner	Analytics Role
page_view	Analytics	Source
lead_form_started	Analytics	Source
inquiry.created	CRM/Core	Consumer
quote.created	Core Commerce	Consumer
quote.accepted	Core Commerce	Consumer
booking.created	Core Commerce	Consumer
booking.status_changed	Core Commerce	Consumer
payment.completed	Payment/Core	Consumer
referral.created	Referral System	Consumer
referral.converted	Referral/Core	Consumer
revenue.recorded	Revenue	Consumer

The ownership matrix prevents multiple systems from becoming competing sources of truth.

16. Service Interest

Primary Categories

- Weddings
- Corporate Events
- Live Shows

- Destination Events

Wedding Categories

- Photography
- Videography
- Decoration
- Makeup
- Catering
- DJ
- Venue
- Entertainment
- Wedding Planning

Example: Photography -> 3 interactions; Decoration -> 2 interactions; Makeup -> 1 interaction. Internal profile: Photography -> HIGH; Decoration -> MEDIUM; Makeup -> LOW.

This is internal analytical intelligence and must not be exposed to vendors/referral partners unnecessarily.

17. Intent Score

Create a configurable 0-100 Intent Score. Initial rules:

Signal	Points
Page view	+1
Service page view	+3
Package/pricing view	+5
Portfolio view	+3
CTA click	+5
Lead form started	+10
Lead completed	+20
Quote requested	+25
Vendor comparison	+10
Booking started	+30
Payment started	+40
Booking completed	+50

Use sensible caps and decay rules where appropriate. Repeated low-value activity must not artificially inflate the score.

Range	Classification
0-19	LOW INTENT
20-39	INTERESTED
40-69	HIGH INTENT
70-100	HOT LEAD

Weights, caps, thresholds, decay and signal rules should be configurable. Future AI scoring should consume the same underlying event data rather than replacing the foundation.

18. Customer Journey Timeline

```
Lead: SV-LEAD-28471

13:02 Website visit
13:03 Wedding page viewed
13:05 Photography page viewed
13:07 Photography package viewed
13:09 Portfolio viewed
13:11 Enquiry started
13:14 Enquiry submitted
13:15 Lead verified
13:20 Vendor matches viewed
13:24 Quote requested

Intent Score: 78
Status: HOT LEAD
Source: Instagram
Referral: SV-REF-XXXX
```

The timeline should clearly distinguish behavioural events from trusted business events.

19. Lead Profile

- Lead ID
- Name
- Phone
- Email
- Event type
- Event date
- Venue/location
- Budget
- Guest count
- Required services
- Lead source
- Campaign
- Referral attribution
- Intent score
- First activity
- Last activity

- Session count
- Enquiry count
- Quote count
- Booking status
- Revenue generated

Access must be role-based and server-authorized.

20. Sales Signals

- HOT LEAD - strong pricing + portfolio + enquiry behaviour
- CALL NOW - high-intent lead with strong contact/quote activity
- FOLLOW-UP - enquiry started but abandoned
- RETURNING VISITOR - visitor returned multiple times
- MULTI-SERVICE INTEREST - multiple service-category engagement
- BOOKING INTENT - booking/payment process started but not completed

These are internal sales intelligence and must never automatically become vendor-facing data.

21. Sales Dashboard

Today's Activity

- Active visitors
- New leads
- Hot leads
- Abandoned forms
- Quotes requested
- Bookings started
- Bookings completed

Funnel

```
Visitors -> Service Interest -> Enquiry -> Verified Lead -> Quote -> Vendor Match -> Booking -> Payment -> Completed Event
```

Display conversion rates between stages.

22. Source / Campaign / Referral Analytics

Source	Referral	Leads	Qualified	Bookings	Booking Value	StarVnt Revenue
Instagram	-	100	15	2	₹50,000	₹7,500
Referral Partner X	Partner X	8	3	1	₹25,000	₹3,750

These examples illustrate the reporting shape only; live values must come from authoritative systems.

23. Revenue Attribution

```
Instagram Campaign A
-> Visitor
-> Lead
-> Wedding Photography
-> Booking ₹25,000
-> StarVnt Revenue ₹3,750
```

```
Referral Partner X
-> Visitor
-> Lead
-> Booking ₹25,000
-> StarVnt Revenue ₹3,750
-> Referral Commission
```

- Leads
- Qualified leads
- Booking conversion
- Gross booking value
- StarVnt revenue
- Revenue per lead
- Revenue per source
- Revenue per campaign
- Revenue per referral
- CAC when ad-spend integration exists
- Referral commission

Financial values must come from authoritative financial systems.

24. Abandoned Lead Recovery

```
Starts enquiry -> Provides partial information -> Does not submit -> FOLLOW_UP_REQUIRED
```

If an identified lead remains inactive for a configurable period, mark FOLLOW_UP_REQUIRED. Do not automatically contact users without appropriate consent/communication permission.

25. Security Foundation

Security must be part of the foundation, but the MVP should avoid unnecessary complexity.

- Existing StarVnt authentication integration
- Server-side RBAC
- Resource-level authorization
- Strict customer/vendor separation
- Secure API authentication
- Schema validation
- Event spoofing protection
- Rate limiting where appropriate
- Secure cookie/local-storage handling
- Consent-aware tracking
- Encryption in transit
- Encryption at rest for sensitive data

- Secure secret management
- Audit logging
- Data retention controls
- Data deletion mechanisms
- Environment separation
- No internal analytics exposed through public APIs
- Idempotency
- Duplicate-event protection
- Security/error monitoring

> *Frontend hiding is not security. Every sensitive operation must be authorized server-side.*

26. Privacy

This must never become a surveillance system. Track only activity inside StarVnt-owned properties for legitimate product, customer-experience, sales, matching, conversion and business-analysis purposes.

- Never track unrelated websites
- Never collect general internet browsing history
- Never use hidden keylogging
- Never record passwords
- Never capture payment-card information
- Never capture private messages
- Never collect unnecessary sensitive data
- Never sell customer behavioural data

Provide appropriate privacy-policy integration, consent mechanism, data-retention controls, data deletion, access controls, audit logs and secure storage.

27. Vendor Privacy

```
Customer -> Wedding Client #SV-2847
Vendor -> Photography Partner #PH-0184
```

- Vendors must never receive intent score
- Vendors must never receive customer journey history
- Vendors must never receive campaign attribution
- Vendors must never receive internal sales intelligence
- Vendors must not receive unnecessary behavioural information

Vendors should receive only the information necessary to fulfil an authorized StarVnt opportunity.

28. Referral Partner Privacy

Referral partners are external users and must be separated from internal Sales/Admin roles.

May receive

- Referral ID
- Referral status

- Approved conversion status
- Approved reward/commission information
- Other explicitly approved referral metrics

Must not receive

- Customer browsing history
- Intent score
- Internal sales notes
- Campaign intelligence
- Private customer information
- Other customers' information
- Other referral partners' information

29. Admin / Access Model

Role	Access
Super Admin	Full analytics + customer journey access
Sales Manager	Lead + funnel + intent access
Sales Executive	Assigned leads only
Vendor Manager	Vendor operational information
Vendor	Assigned/invited opportunities only
Referral Partner	Approved referral information only

```
Frontend -> API -> Authorization -> Resource-level permission -> Data
```

RBAC must be enforced server-side. Never rely only on frontend visibility rules.

30. Technical Architecture

Potential frontend module

```
tracking.ts
```

Potential backend services

```
/analytics/events
/analytics/sessions
/analytics/leads
/analytics/journey
/analytics/intent
/analytics/attribution
/analytics/referrals
```

Potential analytics data domains

```
Visitors
Sessions
Events
LeadEvents
ServiceInterests
```

```
IntentScores
CampaignAttribution
ReferralAttribution
RevenueAttribution
AuditLogs
```

Core Commerce continues to own Customer, Inquiry, Quote, Booking and Payment. Do not duplicate these unless there is a documented cache or integration requirement.

31. Reusable Services

```
trackEvent()
trackPageView()
trackCTA()
trackLeadAction()
trackReferral()
calculateIntentScore()
updateCustomerJourney()
attributeBooking()
attributeRevenue()
```

The implementation should make it easy for future StarVnt properties and products to use the same analytics foundation.

32. Event Reliability

- Unique event_id
- Duplicate detection
- Retry-safe processing
- Idempotent consumers
- Failed-event handling
- Event logging
- Timestamp validation
- Schema versioning

If the same booking.created event is delivered twice, Analytics must not create two bookings or double-count revenue.

33. Failure Handling

The design should define what happens when Analytics, CRM or Core Commerce is unavailable; event delivery fails or is delayed; duplicates arrive; schemas change; referral attribution is missing; a visitor cannot be associated with a lead; or a payment event arrives before a booking event.

> *Analytics failure should never block core customer transactions. Booking and payment must remain functional if Analytics is temporarily unavailable.*

34. Data Retention

Retention should be configurable and data classes may have different requirements.

- Anonymous behavioural events
- Lead activity
- Customer information
- Financial records

- Audit logs

Do not retain behavioural data indefinitely simply because storage is inexpensive. Document retention/deletion policies before production.

35. MVP Roadmap

Phase 1 - Foundation

```
Visitor -> Session -> Page Views -> CTA -> UTM -> Referral Attribution -> Lead Form -> Basic Analytics
```

Phase 2 - Intelligence

```
Journey Timeline -> Service Interest -> Intent Score -> Funnel -> Sales Signals
```

Phase 3 - Commerce

```
Booking Attribution -> Revenue Attribution -> Vendor Integration -> Referral Revenue Attribution
```

Phase 4 - AI

```
Predictive Intent -> AI Lead Scoring -> AI Recommendations -> Conversion Prediction -> Customer Lifetime Value
```

Do not allow Phase 4 requirements to complicate Phase 1 unnecessarily.

36. Source-of-Truth Architecture

Domain	Source of Truth
Visitor behaviour	Analytics
Attribution	Analytics / Attribution Layer
Lead	CRM
Sales activity	CRM
Customer identity	Core Identity / Core Commerce
Inquiry	Core Commerce
Quote	Core Commerce
Booking	Core Commerce
Payment	Payment / Core Commerce
Revenue	Revenue System
Referral relationship	Referral System
Referral commission	Referral System / Revenue
Intent score	Analytics / Intent Layer

No two systems should independently claim authority over the same business domain.

37. Architecture Review Before Coding

Do not start major implementation immediately. First review the existing StarVnt architecture and provide a technical proposal.

1. Technical approach
2. System architecture
3. Existing components that should be reused
4. Required Core Platform changes
5. Required events
6. Event contracts
7. API contracts
8. Database/data-model dependencies
9. Visitor -> Lead identity association
10. Attribution model
11. Referral integration
12. Analytics -> CRM integration
13. CRM -> Core Commerce integration
14. Booking/revenue attribution
15. Authentication approach
16. RBAC model
17. Security model
18. Privacy model
19. Event ownership
20. Source-of-truth mapping
21. Idempotency strategy
22. Retry/failure strategy
23. Data retention strategy
24. Audit requirements
25. Performance/scalability considerations
26. Architectural risks
27. Dependencies on other teams/systems
28. Decisions required from my side

Also identify anything already available in the StarVnt codebase that can be reused instead of rebuilt.

38. Success Criteria

An authorized StarVnt Sales Manager should eventually be able to open a lead and immediately understand:

1. Where the customer came from
2. Whether the source was campaign/referral/organic/etc.
3. What event they are planning
4. Which services they are interested in
5. What they did inside StarVnt
6. Their current intent level
7. Their latest meaningful action
8. Where they dropped off
9. What Sales should do next

10. Whether they booked
11. Booking value
12. StarVnt revenue
13. Referral attribution, if applicable
14. Referral reward/commission, if applicable

> *Customers must not see internal analytics. Vendors must not see internal customer intelligence. Referral partners must not see internal customer intelligence. Analytics must not become surveillance.*

39. Final Architecture Principle

> *Track customer behaviour inside StarVnt, understand intent, preserve acquisition and referral attribution, connect the journey to real commercial outcomes, but keep every domain, role and user's data appropriately isolated.*

The architecture should be: MVP-first + Privacy-conscious + Secure + Production-oriented + Modular + Scalable.

Most importantly, the analytics layer should observe and connect the StarVnt ecosystem, not become the source of truth for every system.

Pre-Implementation Review Package

- Architecture Proposal
- Event Contracts
- API Contracts
- Integration Dependencies
- Security Model
- Source-of-Truth Mapping
- Attribution Strategy
- Major Risks

Once the architecture is reviewed and approved, implementation can proceed phase-by-phase.

Regards,

Subhankar Halder

Founder & CTO | StarVnt Entertainment

Appendix A - Quick Architecture View

```
STARVNT CUSTOMER JOURNEY + INTENT

Acquisition / Referral
|
v
Analytics Layer
Visitor + Session + Events + Attribution
|
v
Revenue CRM
Lead + Qualification + Follow-up + Sales Activity
|
v
Core Commerce
Inquiry + Quote + Booking + Payment
|
v
Revenue
Booking Value + StarVnt Revenue
|
+----> Referral System (approved attribution / commission)

Analytics is a connector + intelligence layer.
Domain systems remain authoritative for their own records.
```

Appendix B - Implementation Guardrails

- Do not build a separate customer identity database when Core already owns identity.
- Do not make browser analytics authoritative for financial events.
- Do not expose intent scores or behavioural history to vendors/referral partners.
- Do not let Analytics downtime block booking or payment.
- Do not create duplicate lead, booking, payment or revenue records.
- Do not over-engineer predictive AI before the event and attribution foundation works end-to-end.
- Do not silently change Core/CRM/Referral ownership boundaries.
- Before adding a major feature, verify which system owns the data, which system emits the event and which system consumes it.

Appendix C - Final North Star

> *The goal is not to collect more data. The goal is to make StarVnt smarter about the customer journey, while remaining privacy-conscious, secure and operationally trustworthy.*

StarVnt should be able to understand the path from acquisition to intent to booking to revenue - without compromising customer trust or turning analytics into surveillance.